

Project Design Phase-II

Data Flow Diagram & User Stories

Date	02 October 2026
Team ID	SWTID-2026-1180
Project Name	Implement Client Script and UI Policy(incidents)
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Real-World Business Scenario

Title: High-Priority Enterprise Incident Logging & Asset Verification

Table: `incident` (or custom Service Desk form)

- **The Situation:** A support agent or employee logs a critical network or server outage on behalf of an executive user.
- **The Failure Points (Without Client Scripts):**
 - The user enters a malformed callback phone number (e.g., letters or fewer than 10 digits).
 - The user sets **Priority** to `1 - Critical` but enters an end-of-life or unassigned Configuration Item (CI/Asset Tag).
 - The system lets the user click **Submit**. Only after sending the request to the database does a server-side business rule reject the record.
 - The page reloads, wipes unsaved notes, forces the agent to re-type diagnostic information, and delays critical issue dispatch.
- **The Solution (With Client Scripts & GlideAjax):**
 - **onLoad:** Pre-populates the current logged-in user profile, direct supervisor, and primary department into caller fields without manual input.
 - **onChange:** The moment the user types the contact number, a client script runs regex validation; if invalid, an inline warning (`showFieldMsg`) appears instantly.

- **onChange (GlideAjax):** When the user selects an Asset/CI, an asynchronous call reaches a Script Include on the server to check CI maintenance status and contract coverage in the background without freezing the browser UI.
- **onSubmit:** If mandatory fields are improperly formatted or contradictory, submission is blocked in the browser (`return false`), keeping all entered notes intact.

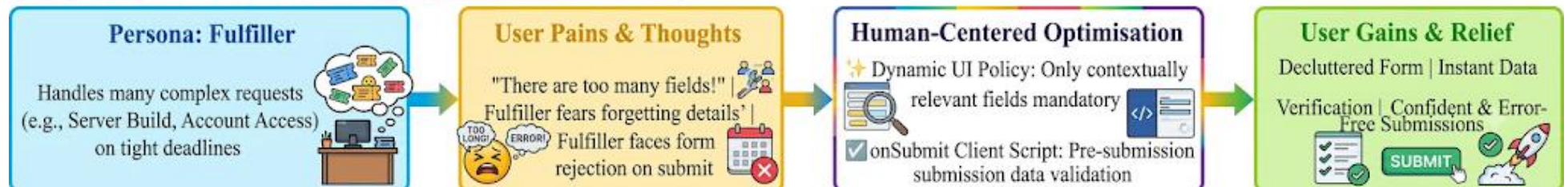
Customer (Web user)	Form Personalization	USN-01	As a web user opening a service form, I want my profile details (name, email, department) to auto-populate so that I do not waste time typing basic info.	• Form auto-fills caller details via <code>onLoad</code> script • Fields remain read-only to avoid accidental overrides	High	Sprint-1
Customer (Web user)	Dynamic Field Visibility	USN-02	As a web user selecting a request category, I want to see only fields relevant to my specific issue so that the form remains uncluttered.	• Selecting 'Hardware' displays asset and serial number fields • Irrelevant software fields remain hidden via UI Policy	High	Sprint-1
Customer (Web user)	Client-Side Format Validation	USN-03	As a web user typing my contact phone number, I want real-time format verification so that I know immediately if the number is invalid.	• <code>onChange</code> script runs regex check on blur/change • Field displays clear inline warning (<code>showFieldMsg</code>) if format is wrong	Medium	Sprint-1

Customer (Web user)	Submission Integrity Check	USN-04	As a web user clicking submit, I want the system to block submission in the browser if required data is invalid so that I do not lose entered data via a page reload.	• <code>onSubmit</code> script halts submission if fields fail validation • Form retains all entered text without reloading	High	Sprint-1
Customer (Mobile user)	Mobile Form Responsiveness	USN-05	As a mobile user logging an incident on my phone, I want mandatory fields highlighted dynamically so that I can complete tickets on small screens quickly.	• UI Policies and Client Scripts execute on Mobile UI Type • Conditional fields show/hide cleanly without horizontal scrolling	Medium	Sprint-2
Customer Care Executive	Dynamic Priority Escalation	USN-06	As a service desk agent updating an incident to 'Critical', I want a mandatory justification field to appear automatically so that review audits are enforced.	• UI Policy activates when Priority changes to '1 - Critical' • Justification field dynamically switches to <code>Mandatory = True</code>	High	Sprint-1
Customer Care Executive	Asynchronous Asset Lookup	USN-07	As a service desk agent selecting a Configuration Item (CI), I want the system to check warranty/status in the background so that the browser does not freeze.	• <code>GlideAjax</code> asynchronously checks CI status via Script Include	Medium	Sprint-2

				• Inactive or retired CI warning displays inline without UI lag		
Customer Care Executive	Read-Only Audit Protection	USN-08	As a support agent viewing a resolved ticket, I want all diagnostic fields locked as read-only so that completed records cannot be accidentally edited.	• UI Policy sets form fields to Read-Only = True when ticket state is 'Resolved' or 'Closed' • Reverse-if-false unchecked	High	Sprint-1
Administrator	Declarative Policy Management	USN-09	As an administrator, I want to manage field visibility rules using declarative UI Policies without writing JavaScript so that maintenance overhead is minimized.	• UI Policy created using condition builder (e.g., Category is Software) • UI Policy Actions manage Visible/Mandatory states cleanly	Medium	Sprint-2
Administrator	Script Performance & Exception Handling	USN-10	As an administrator, I want client scripts to execute asynchronously and handle null values cleanly so that form load times stay under 2 seconds.	• Elimination of synchronous calls like <code>getReference()</code> callbacks • Zero uncaught JavaScript errors in the browser console	Low	Sprint-2

DATA FLOW DIAGRAM:

Empathy-Driven IT Form Optimisation Flow



PROCESS EXPLANATION:

- **1. Persona (Blue Box):**
 - The user (Fulfiller/Employee) accesses the complex request form under strict deadlines.
- **2. Pains & Thoughts (Yellow Box):**
 - The user feels overwhelmed by excessive fields and fears that typing errors will trigger form rejections and data loss.
- **3. Optimisation (Purple Box):**
 - ServiceNow applies dynamic UI Policies to display only relevant fields and `onSubmit` Client Scripts to validate data live in the browser.
- **4. Gains & Relief (Green Box):**
 - The user experiences a clean interface with instant verification, leading to confident, error-free submissions.